



Modelling Financial Markets

Seminar 12: Options and the Volatility Surface — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ATM: At-The-Money
- ▣ CI: Confidence Interval
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DM: Diebold–Mariano (test)
- ▣ DVOL: Deribit Volatility Index
- ▣ FRED: Federal Reserve Economic Data
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ SVIX: Simple VIX: index of simple variance swaps (Martin, 2017)
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index
- ▣ VIX3M: Cboe 3-Month Volatility Index
- ▣ VRP: Variance Risk Premium
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive no-arbitrage restrictions, the Black–Scholes equation, the Greeks and the variance-swap strike, and check them on data
- ▣ Build delta and delta-gamma hedges and measure the hedging error with a confidence interval
- ▣ Fit a smile and a risk-neutral density with bootstrap bands, and test forecasts and the variance premium with robust inference

- ▣ Background: Lecture 12; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations, each checked on numbers (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), an AI answer to critique (C2), and a project seed from the case study of the lecture (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Volatility in annualised %; “vol points” = percentage points of volatility; time in years unless stated
- Overlapping windows are autocorrelated: always use Newey–West standard errors or a block bootstrap [Newey & West \(1987\)](#), [Künsch \(1989\)](#)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (bounds and convexity in the strike), A3 (Black–Scholes equation and hedging P&L), A5 (one identity for the Greeks), A7 (implied volatility: existence and a starting value)
- Part B: B1 (hedging error with bootstrap intervals), B3 (SVI fit of a Bitcoin smile), B5 (variance risk premium of the S&P 500)

Proposed: you solve them

- Part A: A2 (parity as a regression; model: A1), A4 (tree convergence; model: A3), A6 (delta-gamma hedge; model: A5), A8 (variance-swap strike and its errors; model: A1)
- Part B: B2 (delta-hedged S&P 500 options; model: B1, B5), B4 (a VIX for Bitcoin; model: A8, B3), B6 (is the VIX an efficient forecast?; model: B5), B7 (Bitcoin premium; model: B5), B8 (density with bands; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Black–Scholes and parity); C3, is the SVIX bound tight? (case study)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter12_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500 and VIX (1990–2026), VIX3M (2007–2026); daily closes up to 18 September 2026
- Bitcoin: daily closes; DVOL (Deribit, 2021–2026); the Bitcoin option chain of the lecture (Deribit snapshot)
- Two series together: prices joined on common days first, then returns

Formulas Used Today

- Parity: $C - P = Se^{-q\tau} - Ke^{-r\tau}$; binomial: $q = (R - d)/(u - d)$, $C = [qC_u + (1 - q)C_d]/R$ Cox, Ross & Rubinstein (1979)
- Black-Scholes Black & Scholes (1973): $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$
- Greeks: $\Delta = \Phi(d_1)$, $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, vega $= S\varphi(d_1)\sqrt{\tau}$, $\Theta = -S\varphi(d_1)\sigma/(2\sqrt{\tau}) - rKe^{-r\tau}\Phi(d_2)$
- Implied variance Cboe (2025): $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$
- SVI Gatheral & Jacquier (2014): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$, $w = \sigma_{imp}^2 \tau$, $k = \ln(K/F)$
- Variance risk premium Carr & Wu (2009): $VRP_t = IV_t^2 - RV_{t,t+h}$, $RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$ (A = periods per year)

A1: Bounds and Convexity in the Strike [Solved]

□ Context and data

- ▶ European calls on a non-dividend-paying asset, price $C(K)$ for strike K , maturity τ , continuous rate r ; no arbitrage, no frictions
- ▶ Data for (d): Bitcoin calls expiring 30 October 2026 (Deribit snapshot of the lecture, 22 September 2026), BTC prices converted to USD with the Bitcoin index $S = 85\,265$ (the Deribit convention); forward $F = 85\,742$, so $e^{r\tau} = F/S$

□ Tasks

- ▶ (a) Prove $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ by comparing the pay-offs of two portfolios in every state at T
- ▶ (b) For $K_1 < K_2 < K_3$ with $K_2 = \lambda K_1 + (1 - \lambda)K_3$, prove $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$, i.e. C is convex in K
- ▶ (c) Show that the butterfly $[C(K - h) - 2C(K) + C(K + h)]/h^2$ tends to $e^{-r\tau}q(K)$, with q the risk-neutral density [Breedon & Litzenberger \(1978\)](#)
- ▶ (d) Compute the butterfly 84/85/86 thousand USD from mark prices, the implied density, the triangularly weighted mass $e^{r\tau}$ butterfly/ h and $P^Q(84\,000 < S_T < 86\,000)$; then count, over all 732 equally spaced call triples, how many violate convexity at mark prices and at executable prices (wings at the ask, body at the bid)

□ Report and interpret

- ▶ The bounds and the convexity proof; the butterfly value, $q(85\,000)$ and the violation counts; is any violation an arbitrage you could trade?

A1: Solution (1/2) [Solved]

- (a) Lower bound: a call plus $Ke^{-r\tau}$ in cash is worth $\max(S_T, K) \geq S_T$ at T , so it cannot cost less than the share:
 $C + Ke^{-r\tau} \geq S$
 - ▶ The call pays $(S_T - K)^+ \geq 0$, so $C \geq 0$; together $C \geq \max(S - Ke^{-r\tau}, 0)$
 - ▶ Upper bound: the share pays $S_T \geq (S_T - K)^+$ in every state, so $C \leq S$
- (b) For fixed S_T the map $K \mapsto (S_T - K)^+$ is convex, so $\lambda(S_T - K_1)^+ + (1 - \lambda)(S_T - K_3)^+ \geq (S_T - K_2)^+$ in every state
 - ▶ A portfolio that pays at least as much in every state costs at least as much today: $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$
- (c) $C(K) = e^{-r\tau} \int_K^\infty (x - K)q(x) dx$, so $\partial_K C = -e^{-r\tau} \int_K^\infty q(x) dx$ and $\partial_K^2 C = e^{-r\tau} q(K)$
 - ▶ The butterfly quotient is the central second difference, so it converges to $e^{-r\tau} q(K)$: convexity is exactly $q \geq 0$

A1: Solution (2/2) [Solved]

- (d) Mark prices in USD: $C(84\,000) = 4957.26$, $C(85\,000) = 4414.38$, $C(86\,000) = 3922.98$
 - ▶ Butterfly = $4957.26 - 2 \cdot 4414.38 + 3922.98 = 51.47$ USD, positive as required
 - ▶ $e^{r\tau} = F/S = 1.0056$: $q(85\,000) \approx 1.0056 \cdot 51.47 / 1\,000^2 = 5.18 \cdot 10^{-5}$ per USD
 - ▶ The butterfly pays $(h - |S_T - K|)^+$: $e^{r\tau}$ butterfly/ $h = 5.2\%$ is a triangularly weighted mass $\approx h q(K)$; $P^Q(84\,000 < S_T < 86\,000) \approx 2h q(K) = 10.4\%$
- All triples: 7 of 732 mark butterflies are negative, all among deep in-the-money calls, the worst -11.80 USD
 - ▶ The median bid–ask spread at those strikes is 255.80 USD; at executable prices the number of negative butterflies is 0
- Interpretation: no tradable arbitrage; but a density estimated from raw marks would be negative in places, which is why the lecture smooths the smile (SVI) before differentiating

A2: Parity as a Regression: the Implied Forward [Proposed]

Solution discussed in class

□ Context and data

- ▶ European calls and puts with the same expiry and strikes $K_1, \dots, K_n$; discount factor $D = e^{-r\tau}$ and forward F unknown
- ▶ Data: the expiry of A1 (30 October 2026), USD mid prices $(bid + ask)/2 \times S$ (S the Bitcoin index, the Deribit convention), only strikes where both the call and the put have a bid and an ask

□ Tasks

- ▶ (a) Show from put–call parity that $C(K) - P(K) = DF - DK$ for every strike
- ▶ (b) Write the OLS regression $C_i - P_i = a + bK_i + e_i$; derive $\hat{D}$, $\hat{F}$ from $(\hat{a}, \hat{b})$ and the delta-method standard error of $\hat{F}$ with heteroskedasticity-robust covariance
- ▶ (c) Estimate on the data; compare $\hat{F}$ with the exchange forward in standard errors

□ Report and interpret

- ▶ $\hat{D}$, $\hat{F}$ with standard errors, the number of strikes; which USD interest rate does $\hat{D}$ imply, and why might $\hat{F}$ differ from the exchange forward?
- ▶ Model: A1 (portfolios with identical pay-offs)

A3: The Black–Scholes Equation and the Hedging P&L [Solved]

□ Context and data

- ▶ Price $dS_t = \mu S_t dt + \sigma S_t dW_t$, constant r , continuous trading without costs; option value $V(S, t)$, twice differentiable [Black & Scholes \(1973\)](#)
- ▶ Numbers for (c): the call of A5 ($S = K = 100$, six months, $r = 3\%$), sold at $\sigma_{imp} = 25\%$, gamma $\Gamma = 0.0222$; 252 trading days a year

□ Tasks

- ▶ (a) Replicate V with a self-financing portfolio of Δ shares and cash B at r ($d\Pi = \Delta dS + rB dt$); apply Itô's lemma to V , choose Δ to remove the dW_t term and use no arbitrage to derive the Black–Scholes equation
- ▶ (b) A dealer sells the option at σ_{imp} , holds $\Delta = \partial_S V^{imp}$ shares and cash at r , while S moves with σ_{real} ; derive the instantaneous P&L
- ▶ (c) Compute today's P&L rate per trading day if $\sigma_{real} = 30\%$, then annualise it with S and Γ frozen at today's values

□ Report and interpret

- ▶ The equation, the P&L formula and the two numbers; who gains and why does the P&L depend on the path of S ?

A3: Solution (1/2) [Solved]

- (a) Itô: $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$; the self-financing $\Pi = \Delta S + B$, started at $\Pi = V$, has $d\Pi = \Delta dS + rB dt$
- $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt + (V_S - \Delta)(\mu S dt + \sigma S dW_t)$; with $\Delta = V_S$ both the risk and μ disappear
- A riskless position that costs 0 earns 0: $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = rB = r(V - SV_S)$, i.e. $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
- (b) The self-financing book $-V + \Delta S + (V - \Delta S)$ in cash changes by $-dV + \Delta dS + r(V - \Delta S) dt$, plus interest on the P&L already accumulated
- Itô with the true volatility: $dV = V_t dt + \Delta dS + \frac{1}{2}\Gamma \sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2}\sigma_{imp}^2 S^2 \Gamma$

A3: Solution (2/2) [Solved]

- Substituting, the dS and r terms cancel: $d \text{ P\&L} = r \text{ P\&L} dt + \frac{1}{2} \Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$; the second term is the excess gain
- (c) $\frac{1}{2} \cdot 0.02223 \cdot 100^2 \cdot (0.25^2 - 0.30^2) \cdot \frac{1}{252} = \frac{1}{2} \cdot 0.02223 \cdot 10^4 \cdot (-0.0275)/252 = -0.0121$ per option and trading day
- Annualised with S and Γ frozen: -3.06 , about 39% of the call price 7.76; a rate, not the realised P&L of this six-month option, since ΓS^2 changes along the path
- Interpretation: the seller loses when realised volatility exceeds implied; the loss accrues through Γ , largest near the strike and near expiry, so two paths with the same realised variance give different P&L

A4: The Binomial Tree Converges to Black–Scholes [Proposed]

Solution discussed in class

□ Context and data

- ▶ Cox–Ross–Rubinstein (CRR) tree with N steps of length $\Delta t = T/N$ Cox, Ross & Rubinstein (1979); European call of the lecture: $S = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$

□ Tasks

- ▶ (a) Show that $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $q = (e^{r\Delta t} - d)/(u - d)$ match the risk-neutral mean exactly and the variance of $\ln(S_{t+\Delta t}/S_t)$ up to terms of order Δt^2
- ▶ (b) Price the call with $N = 50, 51, 100, 101, 200, 201$ steps and compute $N(C_N - C_{BS})$ for each
- ▶ (c) Average the prices for $N = 200$ and $N = 201$ and compare the error with that of $N = 200$

□ Report and interpret

- ▶ A table of C_N and $N(C_N - C_{BS})$; the order of the error, the reason for the even/odd oscillation, and why averaging helps
- ▶ Model: A3 (from the tree to the continuous-time equation)

A5: One Identity for All the Greeks [Solved]

□ Context and data

- ▶ Black–Scholes call $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$; φ = standard Normal density
- ▶ Numbers for (c): $S = K = 100$, six months ($\tau = 0.5$), $r = 3\%$, $\sigma = 25\%$, no dividends

□ Tasks

- ▶ (a) Prove $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- ▶ (b) Use (a) to derive $\Delta = \partial_S C = \Phi(d_1)$ and $\text{vega} = \partial_\sigma C = S\varphi(d_1)\sqrt{\tau}$; then show $\text{vega} = \Gamma S^2\sigma\tau$ with $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$
- ▶ (c) Compute d_1 , d_2 , both sides of (a), Δ , Γ , vega per unit of σ in two ways, the call and the put (from parity)

□ Report and interpret

- ▶ The proofs and the numerical checks; what does $\text{vega} = \Gamma S^2\sigma\tau$ imply for hedging gamma with options of the same expiry?

A5: Solution (1/2) [Solved]

- (a) $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2}$ and $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau} \cdot \frac{2 \ln(S/K) + 2r\tau}{\sigma\sqrt{\tau}}$
- Hence $\varphi(d_2)/\varphi(d_1) = Se^{r\tau}/K$, i.e. $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- (b) $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)] \partial_S d_1 = \Phi(d_1)$, since $\partial_S d_1 = \partial_S d_2$ and the bracket is 0 by (a)
- Vega: $S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
- And $\Gamma S^2 \sigma \tau = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} S^2 \sigma \tau = S\varphi(d_1)\sqrt{\tau} = \text{vega}$

A5: Solution (2/2) [Solved]

- (c) $\sigma\sqrt{\tau} = 0.1768$; $d_1 = (0 + (0.03 + 0.03125) \cdot 0.5)/0.1768 = 0.1732$, $d_2 = -0.0035$
- Identity: $S\varphi(d_1) = 100 \cdot 0.3930 = 39.30$ and $Ke^{-r\tau}\varphi(d_2) = 98.51 \cdot 0.3989 = 39.30$: equal
- $\Delta = \Phi(0.1732) = 0.5688$; $\Gamma = 0.3930/(100 \cdot 0.1768) = 0.0222$
- $\text{Vega} = 100 \cdot 0.3930 \cdot \sqrt{0.5} = 27.79$ and $\Gamma S^2 \sigma \tau = 0.02223 \cdot 10^4 \cdot 0.25 \cdot 0.5 = 27.79$ per unit of σ (0.2779 per vol point)
- $C = 100 \cdot 0.5688 - 98.51 \cdot 0.4986 = 7.76$; parity: $P = C - 100 + 98.51 = 6.27$
- Interpretation: for one expiry, vega and gamma are proportional, so a gamma-neutral book of same-expiry options is also vega-neutral, but only while the smile is flat (A6)

A6: A Delta-Gamma Hedge [Proposed]

Solution discussed in class

□ Context and data

- ▶ A dealer has **sold** 1 000 calls of A5 ($S = K = 100$, six months, $r = 3\%$, $\sigma = 25\%$)
- ▶ Hedging instruments: a call with $K = 110$, same expiry, Black–Scholes price 3.90, and the stock; Greeks from the Black–Scholes formulas

□ Tasks

- ▶ (a) Compute the delta and the gamma of the dealer's position
- ▶ (b) Find the number n_2 of $K = 110$ calls that makes the gamma zero, then the number of shares that makes the delta zero
- ▶ (c) Compute the vega of the hedged book and explain the result

□ Report and interpret

- ▶ The positions in the two instruments and the residual vega; is the book protected against a change in volatility?
- ▶ Model: A5

A7: Implied Volatility: Existence, Uniqueness, Starting Value [Solved]

□ Context and data

- ▶ European call, Black–Scholes price $BS(\sigma)$; market price 4.50 for $S = K = 100$, three months ($\tau = 0.25$), $r = 2\%$

□ Tasks

- ▶ (a) Prove that $\sigma \mapsto BS(\sigma)$ is continuous and strictly increasing, with limits $(S - Ke^{-r\tau})^+$ as $\sigma \rightarrow 0$ and S as $\sigma \rightarrow \infty$; conclude when an implied volatility exists
- ▶ (b) Derive the at-the-money-forward approximation $C \approx S\sigma\sqrt{\tau}/\sqrt{2\pi}$ [Brenner & Subrahmanyam \(1988\)](#)
- ▶ (c) Use (b), with and without subtracting half the intrinsic value $(S - Ke^{-r\tau})/2$, as a starting value; then run one Newton step $\sigma_1 = \sigma_0 - (BS(\sigma_0) - 4.50)/\mathcal{V}(\sigma_0)$ from $\sigma_0 = 20\%$

□ Report and interpret

- ▶ The two starting values, σ_1 and the exact implied volatility; why does Newton converge so fast here, and when would it fail?

A7: Solution (1/2) [Solved]

- (a) $\partial_{\sigma} BS = S\varphi(d_1)\sqrt{\tau} > 0$ (A5): strictly increasing and continuous
- $\sigma \rightarrow 0$: $d_{1,2} \rightarrow \pm\infty$ according to the sign of $\ln(S/K) + r\tau$, so $BS \rightarrow (S - Ke^{-r\tau})^+$; $\sigma \rightarrow \infty$: $d_1 \rightarrow \infty$, $d_2 \rightarrow -\infty$, so $BS \rightarrow S$
- By the intermediate value theorem a unique σ_{imp} exists iff the price lies strictly inside the no-arbitrage bounds of A1
- (b) If $S = Ke^{-r\tau}$: $d_{1,2} = \pm\sigma\sqrt{\tau}/2$ and $C = S[\Phi(\sigma\sqrt{\tau}/2) - \Phi(-\sigma\sqrt{\tau}/2)] \approx S\sigma\sqrt{\tau}\varphi(0) = S\sigma\sqrt{\tau}/\sqrt{2\pi}$

A7: Solution (2/2) [Solved]

- (c) Without correction: $\sigma \approx 4.50\sqrt{2\pi}/(100 \cdot 0.5) = 22.56\%$
- With correction: $Ke^{-r\tau} = 99.501$, half the intrinsic value 0.249, $\sigma \approx (4.50 - 0.249)\sqrt{2\pi}/50 = 21.31\%$
- Newton: $BS(20\%) = 4.2322$, $\mathcal{V} = 19.848$, so $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$; exact 21.3495%
- Interpretation: near the money the price is almost linear in σ , so one Newton step gives four decimals, and the corrected approximation is already within 0.05 vol points
- Deep out of the money vega is close to 0 and Newton can overshoot; there, use a bracketing solver (Brent) inside the bounds

A8: The Variance-Swap Strike and Its Estimation Error [Proposed]

Solution discussed in class

□ Context and data

- ▶ Spanning formula of the lecture: $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$
Bakshi, Kapadia & Madan (2003)
- ▶ Log-normal benchmark: $F = 100$, $r = 0$, 90 days ($T = 90/365$), flat volatility 20%, so the exact variance is 0.04; OTM prices from Black–Scholes

□ Tasks

- ▶ (a) Apply the formula with $f(S) = -2 \ln(S/F)$ and show $K_{var} = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$ for continuous paths
- ▶ (b) Compute the Cboe sum with $\Delta K = 5$ on $[80, 120]$ Cboe (2025), a dense integral on $[80, 120]$ and a dense integral on $(0, \infty)$
- ▶ (c) Split the total error into discretisation (Cboe minus dense on $[80, 120]$) and truncation (dense on $[80, 120]$ minus exact)

□ Report and interpret

- ▶ The three variances and volatilities and the two errors; which dominates here, and what changes for a longer maturity?
- ▶ Model: A1 (portfolios of options as prices of pay-off shapes)

B1: Hedging Error and Rebalancing Frequency [Solved]

▣ Context and data

- ▶ Simulated data: 10 000 paths of geometric Brownian motion, $S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, 252 steps over three months (seed 42)
- ▶ A three-month ATM call ($K = 100$, $r = 4\%$) is sold at its Black–Scholes price and delta-hedged at $\sigma = 20\%$

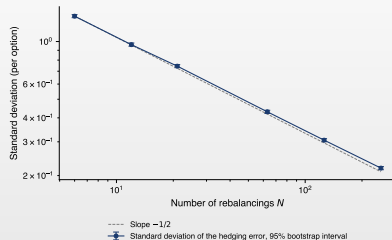
▣ Tasks

- ▶ (a) Rebalance $N = 6, 12, 21, 63, 126, 252$ times; compute the standard deviation of the hedging error at expiry for each N
- ▶ (b) Build 95% percentile bootstrap intervals for each standard deviation from 1 000 resamples of the paths [Efron \(1979\)](#)
- ▶ (c) Regress $\ln(\text{s.d.})$ on $\ln N$; give the slope with a bootstrap interval (the same 1 000 resamples)

▣ Report and interpret

- ▶ A table of s.d. and intervals and a log-log chart; is the slope consistent with $-1/2$? How often must a dealer rebalance to keep the error below 10% of the premium? What would transaction costs change?

B1: Solution (1/2) [Solved]



N	6	12	21	63	126	252
s.d.	1.356	0.963	0.744	0.430	0.306	0.219
95% CI	[1.331, 1.381]	[0.945, 0.983]	[0.731, 0.758]	[0.422, 0.438]	[0.300, 0.312]	[0.215, 0.223]

B1: Solution (2/2) [Solved]

- Step 1: for each path, sell the call at the premium 4.49, buy Δ_0 shares, finance with cash at r ; at each rebalancing date reset Δ ; at expiry, error = shares + cash - pay-off
- Step 2: the s.d. falls from 1.356 ($N = 6$) to 0.219 ($N = 252$); the intervals are narrow because 10 000 paths give a precise s.d.
- Step 3: slope -0.489 , 95% CI $[-0.496, -0.482]$: close to $-1/2$, slightly above it at these finite N (the $1/\sqrt{N}$ law is asymptotic)
- Reading: daily hedging leaves 9.6% of the premium, about the 10% target; four times a day halves it again (ratio 1.97)
- With a fixed fee per trade, costs grow like N while the error falls like $1/\sqrt{N}$; the optimal frequency depends on the cost model and on the chosen trade-off between risk and cost

B2: Delta-Hedged S&P 500 Options, 1990–2026 [Proposed]

Solution discussed in class

□ Context and data

- ▶ Daily S&P 500 closes and VIX (GSPC.INDX, VIX.INDX in data/market), January 1990 to September 2026; interest and dividends ignored
- ▶ Every 21 trading days, sell a one-month ATM call priced with the VIX and hedge it daily at the same volatility (the stylised back-test of the lecture: VIX, not ATM quotes) [Bakshi & Kapadia \(2003\)](#); P&L in % of the index

□ Tasks

- ▶ (a) Compute the mean monthly P&L and its Newey–West standard error with 3 lags; test H_0 : mean = 0 at the 5% level
- ▶ (b) Build a 95% moving-block bootstrap interval for the mean (blocks of 6 months, 2 000 resamples) [Künsch \(1989\)](#)
- ▶ (c) Compute the share of positive months, the skewness and the share of the total gain lost in the five worst months

□ Report and interpret

- ▶ The mean with both intervals and the tail statistics; is the premium significant? Why can a strategy with a high t -statistic still be dangerous? What would interest and dividends change?
- ▶ Model: B1 (bootstrap), B5 (Newey–West)

B3: An SVI Smile for Bitcoin [Solved]

□ Context and data

- ▶ Deribit snapshot of the lecture (`ch12_deribit_snapshot.csv`); expiry closest to 30 days: 2026-10-30, 38 days, 56 quoted OTM options (puts below the forward, calls above)
- ▶ Total variance $w(k) = \sigma_{\text{mark}}^2 \tau$ at log-moneyness $k = \ln(K/F)$; SVI $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + s^2})$ Gatheral & Jacquier (2014)

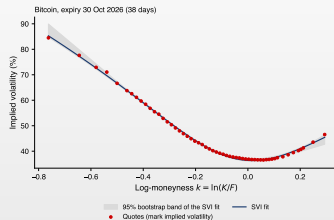
□ Tasks

- ▶ (a) Fit the five SVI parameters by least squares on w , with $b \geq 0$, $|\rho| < 1$, $s > 0$; report the fit error in vol points
- ▶ (b) Pairs bootstrap (300 resamples of the quotes, refit each time): 95% intervals for the ATM volatility and the skew $IV(0.15) - IV(-0.15)$
- ▶ (c) Evaluate the butterfly function $g(k)$ of Gatheral & Jacquier (2014) on $k \in [-1.5, 1.5]$ and report its minimum

□ Report and interpret

- ▶ Parameters, intervals and $\min g$; which parameter controls the skew and what is its sign? Is the skew significantly different from 0? Why fit w and not the prices?

B3: Solution (1/2) [Solved]



- (a) $\hat{a} = -0.020$, $\hat{b} = 0.102$, $\hat{\rho} = -0.262$, $\hat{m} = -0.079$, $\hat{s} = 0.348$; root mean squared error 0.42 vol points

B3: Solution (2/2) [Solved]

- (b) ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew -2.11 , 95% CI $[-2.73, -1.52]$
- ρ controls the tilt: $\hat{\rho} < 0$ makes the put wing dearer; the skew interval excludes 0, so the tilt is significant at 5%
- (c) $\min_k g(k) = 0.136 > 0$: no butterfly arbitrage in this slice, so the implied density is positive
- Why w : it is comparable across strikes and nearly linear in the wings, while option prices span four orders of magnitude and would weight the fit towards the dearest options
- Caveat: the bootstrap resamples smoothed mark volatilities, so the intervals measure fitting noise, not bid–ask uncertainty

B4: A VIX for Bitcoin [Proposed]

Solution discussed in class

□ Context and data

- ▶ Same snapshot; the two expiries around 30 days (2026-10-09, 2026-10-30); USD mid price $Q = \text{BTC mid} \times \text{index } S$ (the Deribit convention); F and $e^{RT} = F/S$ from the Deribit forward

□ Tasks

- ▶ (a) Compute the implied variance of each expiry with the Cboe formula [Cboe \(2025\)](#): K_0 = first strike at or below F ; puts below K_0 , calls above, the put–call average at K_0 ; no zero bids, each wing stops after two consecutive zero bids; ΔK_i = half the distance between neighbouring strikes, one-sided at the ends
- ▶ (b) Interpolate the total variances linearly in time to 30 days and take $100\sigma_{30}$
- ▶ (c) Compare with the ATM volatilities of the two expiries and with DVOL at the same moment [Alexander & Imeraj \(2020\)](#)

□ Report and interpret

- ▶ The two variances, the weight of the near expiry and the index; why is the index above both ATM volatilities, and which choices (strike cut-off, mid vs mark prices) could explain the gap to DVOL?
- ▶ Model: A8 (strip formula and its errors), B3 (the same snapshot)

B5: The Variance Risk Premium of the S&P 500 [Solved]

□ Context and data

- ▶ Daily S&P 500 and VIX closes, joined on common days, January 1990 to August 2026 (last window ending in September 2026)
- ▶ Ex-post premium $VRP_t = VIX_t^2 - RV_{t,t+21}$, $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ in %² Carr & Wu (2009); vol points: $VIX_t - \sqrt{RV_{t,t+21}}$

□ Tasks

- ▶ (a) Compute the mean VRP, its Newey–West standard error with 21 lags Newey & West (1987), a 95% interval and the naive standard error $s/\sqrt{n}$
- ▶ (b) Repeat in vol points and compute the share of days with a positive premium
- ▶ (c) Repeat for 1990–2007 and 2008–2026, test each subperiod mean at the 5% level, then test their equality: regress VRP_t on a constant and a 2008–2026 dummy (Newey–West, 21 lags)

□ Report and interpret

- ▶ A table by period; why is the naive standard error wrong here, and by how much? Is the premium stable across subperiods? Why is the variance premium less precise than the vol-point premium?

B5: Solution (1/2) [Solved]

	n	Mean VRP (% ²)	95% CI	Vol points (s.e.)	Positive
1990–2026	9 203	113.1	[72.7, 153.6]	4.09 (0.27)	86%
1990–2007	4 516	149.3	[125.2, 173.5]	4.60 (0.25)	88%
2008–2026	4 687	78.3	[2.7, 153.9]	3.60 (0.46)	84%

- Step 1: build $RV_{t,t+21}$ from the next 21 log returns, then VRP_t on every day with a complete window
- Step 2: regress VRP_t on a constant with HAC (Bartlett, 21 lags: the overlap; check longer bandwidths) standard errors; interval = mean ± 1.96 s.e.

B5: Solution (2/2) [Solved]

- (a) Full sample: $t = 113.1/20.6 = 5.49$: reject H_0 at 5%; naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days
- (b) In vol points the premium is 4.09 (s.e. 0.27), positive on 86% of days
- (c) 1990–2007: $t = 12.12$; 2008–2026: $t = 2.03$: both reject at 5%, but after 2008 barely, because of the crisis windows of 2008 and 2020
- Equality: difference $-71.0\% ^2$ (s.e. 40.5, $t = -1.76$, $p = 0.08$); in vol points -1.01 ($t = -1.92$, $p = 0.06$)
- Reading: squaring magnifies the crisis windows, so the variance premium has heavier tails and a wider interval than the vol-point premium; the premium is positive in both subperiods, but the difference in size is not significant at 5%: the data cannot tell whether it changed

B6: Is the VIX an Efficient Forecast? [Proposed]

Solution discussed in class

□ Context and data

- ▶ The daily series of B5: $RV_{t,t+21}$ and VIX_t^2 , 1990–2026; for (d) also daily squared returns to build HAR-RV regressors (1, 5 and 22-day averages) [Corsi \(2009\)](#)

□ Tasks

- ▶ (a) Estimate $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ [Mincer & Zarnowitz \(1969\)](#) with Newey–West errors (21 lags); test $\alpha = 0$ and $\beta = 1$ separately and jointly (Wald, 5%); report $\text{corr}(\hat{\alpha}, \hat{\beta})$
- ▶ (b) Add $RV_{t-21,t}$; then re-estimate the regression in logs, $\ln RV$ on $\ln VIX^2$, and test $\beta = 1$
- ▶ (c) Instrument $\ln VIX_t^2$ with $\ln VIX_{t-21}^2$ [Christensen & Prabhala \(1998\)](#) (just-identified GMM, HAC covariance); report the first-stage t and test $\beta = 1$
- ▶ (d) From 2000, re-estimate every 21 days on targets already observed; compare VIX^2 with HAR-RV by QLIKE loss and a Diebold–Mariano test with HAC (21 lags) [Diebold & Mariano \(1995\)](#), [Patton \(2011\)](#)

□ Report and interpret

- ▶ All estimates and tests with decisions at 5%; why can both t -statistics be small while the joint test rejects? How do the conclusions change in logs and with instruments, and what must the instrument satisfy? Does in-sample significance survive out of sample?
- ▶ Model: B5 (Newey–West inference)

B7: The Variance Risk Premium of Bitcoin [Proposed]

Solution discussed in class

□ Context and data

- ▶ DVOL (Deribit), March 2021 to August 2026, and daily Bitcoin closes (BTC-USD.CC); realised volatility over the next 30 calendar days, $A = 365$

□ Tasks

- ▶ (a) Compute the mean premium $DVOL_t - \sqrt{RV_{t,t+30}}$ in vol points, its Newey–West standard error with 30 lags, and test it at 5%
- ▶ (b) Compute the share of positive days and the ratio mean implied / mean realised volatility; compare both with the S&P 500 of B5
- ▶ (c) Count the non-overlapping 30-day periods in the sample

□ Report and interpret

- ▶ Premium, test, ratios and effective sample size; is the Bitcoin premium larger or smaller than the equity premium, and in which unit? How much can five years of data tell us?
- ▶ Model: B5

B8: A Risk-Neutral Density with Confidence Bands [Proposed]

Solution discussed in class

□ Context and data

- ▶ Deribit snapshot of the lecture; expiry closest to 90 days: 2026-12-25, 94 days, 58 OTM quotes; SVI fitted as in B3; forward (undiscounted) prices

□ Tasks

- ▶ (a) Compute the density $q(K) = \partial^2 C / \partial K^2$ from the SVI call prices (Black-76 with F) on a fine strike grid [Breedon & Litzenberger \(1978\)](#)
- ▶ (b) Compute $P^Q(S_T < 0.8F)$ and $P^Q(S_T > 1.2F)$ with 95% pairs-bootstrap intervals (300 refits) and the log-normal values at the ATM volatility
- ▶ (c) Report where the quotes end in K/F and which part of each probability relies on extrapolated wings

□ Report and interpret

- ▶ The two probabilities with intervals vs log-normal; is the crash probability significantly above or below the log-normal one? What does the bootstrap band not measure?
- ▶ Model: B3 (SVI fit and pairs bootstrap)

C1: Is the Variance Risk Premium a Tradable Signal?

Question

- Would you sell variance on the S&P 500 or on Bitcoin every month, and when would you stop? Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- Instrument: a synthetic variance swap with vega notional 1, $P\&L = (IV^2 - RV)/(2 IV)$ vol points per month
- S&P 500 with the VIX (1990–2026); Bitcoin with DVOL (2021–2026)

What a good answer contains

- Sharpe ratio with a block-bootstrap interval, skewness, worst month, maximum drawdown
- A timing rule, e.g. skip months that start in backwardation (VIX above VIX3M); is it better out of sample?
- Comparison with the delta-hedged option of the lecture; position sizing and margin
- The ex-ante VRP ($VIX_t^2/12 - RV_t$) as a predictor of 1–12-month excess returns, with Hodrick (1992) errors and a bootstrap under the null Bollerslev, Tauchen & Zhou (2009), Hodrick (1992)

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Price a European call with 90 calendar days to expiry on a non-dividend-paying stock with $S = 100$, $K = 105$, volatility 20%, continuous rate $r = 4\%$; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)             # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains **three** errors; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a no-arbitrage bound, a unit check, the lecture, a second method
 - ▶ give the corrected line
- Run the corrected code; report C and P and check put-call parity numerically
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Before running the code: what should a 90-day call 5% out of the money cost, roughly: 0.2, 2 or 20?
- Which error would still give a plausible-looking price? Which check catches it?
- Why is “the put comes from parity” not a check of the call price?

C3: Is the SVIX Bound on the Equity Premium Tight? (1/2) [Proposed]

Solution discussed in class

□ Context and data

- ▶ Lecture 12, case study: [Martin \(2017\)](#) shows that $\frac{1}{\tau}(E_t R_T - R_{f,t}) \geq R_{f,t} \text{SVIX}_{t \rightarrow T}^2$ under the negative correlation condition, $\tau = T - t$ in years
- ▶ Data: the author's daily SVIX² series for horizons of 1, 2, 3, 6 and 12 months, 4 January 1996 – 31 January 2012 ([author's data page](#)); S&P 500 daily closes (GSPC.INDX in data/market); 3-month Treasury bill rate (FRED DTB3, Board of Governors of the Federal Reserve System)
- ▶ For (e): the Bitcoin option snapshot of the lecture (ch12_deribit_snapshot.csv), the expiry closest to 30 days, as in B3 and B4

□ Tasks

- ▶ (a) Convert the bill rate into a gross riskless return $R_{f,t}$ over each horizon, compute the bound $R_{f,t} \text{SVIX}_{t \rightarrow T}^2$ in annualised %, and reproduce the mean, standard deviation, median and maximum of Table I of the paper
- ▶ (b) For each horizon, compute the annualised excess return $\frac{1}{\tau}(R_T - R_{f,t})$ from every trading day t , estimate regression (16) by least squares, and compute Hansen–Hodrick standard errors [Hansen & Hodrick \(1980\)](#) with as many lags as there are trading days in the horizon, minus one
- ▶ (c) Test $H_0: \alpha = 0, \beta = 1$ jointly with a Wald test built on the same covariance matrix, then repeat (b) and (c) without the windows that overlap 1 August 2008 – 31 July 2009, as in Table A.1

C3: Is the SVIX Bound on the Equity Premium Tight? (2/2) [Proposed]

Solution discussed in class

Tasks

- ▶ (d) Impose $\alpha = 0$, $\beta = 1$ and compute $R_{OS}^2 = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$ against the historical mean excess return known at t [Welch & Goyal \(2008\)](#); the paper starts that mean in 1871, the course data in 1990: state this as a data limitation
- ▶ (e) Compute Bitcoin SVIX² with eq. (12) of the paper from the OTM options of the snapshot (the same options as the VIX-type index of B4), and compare 100 SVIX with that index

Report and interpret

- ▶ A table with your statistics next to Tables I, II and A.1 of the paper, the Wald p -values and the naive least-squares standard errors; explain every gap larger than rounding
- ▶ Interpretation: does “ $\beta = 1$ is not rejected” support a tight bound, or only reflect low power? Is the gap between Bitcoin SVIX and its VIX-type index consistent with Result 3, and could $R_{f,t}$ SVIX² be a lower bound for Bitcoin?
- ▶ Model: B5 (regression with overlapping windows and HAC errors), B3 (the Bitcoin snapshot)

Common Pitfalls

- ▣ Forgetting dividends or the interest rate in parity: a false arbitrage
- ▣ Treating the risk-neutral probability as a forecast
- ▣ Quoting vega or theta without the unit (per vol point? per day? per year?)
- ▣ Newton's method on deep OTM options ($\text{vega} \approx 0$): use a bracketing solver
- ▣ Naive standard errors on overlapping windows: t -statistics several times too large
- ▣ Reading a high Sharpe ratio of a short-volatility strategy without its skewness and worst month

Take-Home Assignment (Team Project)

- ▣ Take a new Deribit snapshot for Ether (ETH) options and build the smile of three expiries with SVI, with bootstrap intervals
- ▣ Compute a 30-day VIX-type index and the risk-neutral density; compare with Bitcoin [Pele et al. \(2023\)](#)
- ▣ Delta-hedge a sold one-week ATM Bitcoin option on historical paths since March 2021 [Matic, Packham & Härdle \(2023\)](#)
 - ▶ Price and hedge each option at the DVOL of its trade date (the information then available), $r = 0$, daily rebalancing at the 00:00 UTC closes, non-overlapping weekly windows
 - ▶ Report the mean and standard deviation of the error in % of the premium, with moving-block bootstrap intervals; justify every choice you change
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

-  XFG Advanced Methods in Quantitative Finance
-  Pricing Kernels and Risk Premia
-  Hedging Cryptocurrency Options

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12



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